

IT PROJECT MANAGEMENT | 21MGH303P

An AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms and Analyzing Zero's Foundational Role in Modern Binary Computing and Digital Architecture

Project Manager : Ashray Singh

Developer : Madhav Nehra | AI & Content Engineer : Rudransh Yadav

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Unit 1: Introduction to Project Management

Project Manager: Ashray Singh | Developer: Madhav Nehra | AI Engineer: Rudransh Yadav

Experiments Covered in This Unit

Exp 1.1: Project Requirement Gathering and Analysis

Exp 1.2: Project Identification, Methodology and Stakeholder Description

Exp 1.3: Project Cost Estimation and Capital Budgeting

Exp 1.4: Market Demand Analysis and Demand Planning

Problem Statement & Project Objective

Why This Project?

- No existing platform combines AI narration with interactive ancient Indian mathematics
- Students lack engaging, interactive tools to learn Indian numeration history
- The link between Indian Zero (Sunya) and modern binary/digital computing is poorly taught
- Vedic mathematics and abacus-based learning tools remain inaccessible in digital form
- National Education Policy (NEP) 2020 calls for heritage-integrated STEM education

Project Objective

Build an AI-powered educational web app that teaches the evolution of the Indian numeration system, the role of Zero, and its impact on modern digital computing through interactive simulations and Gemini-driven story narration.

Business Case & Project Justification

Why Approve This Project?

- EdTech India market projected at USD 10.4 Bn by 2025 (CAGR 16.5%)
- NEP 2020 mandates heritage + STEM integration in curriculum
- No competing platform offers AI + ancient Indian math simulation
- Target: 25,000+ schools and 4,500+ engineering colleges in India
- Revenue model: Institutional licensing INR 0.75 Lakhs/year

- Total Estimated Budget: INR 10 Lakhs
- Expected NPV (10% discount): +INR 28.38 Lakhs
- Payback Period: ~14 months
- IRR: ~78% (>> 10% hurdle rate)
- Profitability Index: 3.84 (> 1 Accept)

What We Are Delivering

IN SCOPE

- AI Story Narration Module (Gemini LLM)
- Interactive Abacus Simulator
- Vedic Math & Ancient Calculation Methods
- Number System Explorer (Dec/Bin/Oct/Hex)
- Zero & Binary Connection Module
- Quiz & Self-Assessment Module
- Firebase Auth (Student / Educator roles)
- Vercel Deployment + GitHub CI/CD

OUT OF SCOPE

- Native mobile app (iOS / Android)
- Real-time live tutoring / video features
- Third-party LMS integration
- OCR / image recognition features
- Content beyond defined curriculum modules

Stakeholder Analysis

Who Matters to This Project?

Stakeholder	Role	Power	Interest	Engagement
Faculty Advisor	Academic oversight & grading	High	High	Manage Closely
Ashray Singh (PM)	Planning, scheduling, reporting	High	High	Manage Closely
Madhav Nehra	Frontend & Backend Dev	Medium	High	Keep Informed
Rudransh Yadav	AI & Content Engineering	Medium	High	Keep Informed
Students (Users)	Primary learners	Low	High	Keep Informed
Educators	Content managers	Low	Medium	Monitor
Institution / College	Curriculum compliance	High	Medium	Keep Satisfied

Requirements Overview

What Must the System Do?

Functional Requirements (Must-Have)

- FR01: AI story narration using Gemini LLM
- FR02: Interactive abacus simulator
- FR03: Ancient Vedic Math calculation module
- FR04: Number system explorer (Dec/Bin/Oct/Hex)
- FR05: Zero & binary architecture connection module
- FR06: Quiz and student self-assessment module
- FR07: Firebase Auth (role-based access)
- FR08: Admin & content management dashboard

Non-Functional Requirements

- Performance: Page load < 3 seconds
- AI Response Time: < 5 seconds (warm)
- Scalability: 500 concurrent users
- Availability: 99% uptime
- Security: Firebase role-based access
- Usability: Student onboarding < 30 min
- Compatibility: Chrome, Firefox, Edge

Feasibility, Cost & Market Potential

Can We Build It? Should We?

Estimated Budget

INR 10 Lakhs

NPV (10% rate)

+INR 28.38 L

Payback Period

~14 Months

IRR

~78%

Feasibility Summary

- Technical: Next.js + Firebase + Gemini - fully feasible
- Economic: Positive NPV; PI = 3.84
- Operational: 3-member team; all tools accessible
- Risk: Gemini API latency - mitigated with caching

Market Opportunity

- Year 1: Pilot - 5 institutions (free)
- Year 2: 10 paid institutions = INR 7.5 L
- Year 3: 25 institutions = INR 22.75 L
- Year 5: 80 institutions = INR 72 L/yr